

MUHAMMAD RAFAY IRFAN

Software Engineer · Python · C++ · AWS · AI Agents & LLM Systems · End-to-End Product Delivery

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WORK EXPERIENCE

Inkscape / lib2geom (Google Summer of Code)

May 2025 – Sep 2025

Open Source Developer, Remote

- Owned a geometry-library deliverable end-to-end in C++: scoped the problem, selected the numerical methods, and shipped a 50+ case Google Test suite covering tangent, overlapping, and precision-sensitive edge cases, merged into a production codebase used by millions.
- Wrote high-quality, maintainable C++ to resolve a long-standing robustness gap in lib2geom's intersection engine, replacing fragile floating-point logic with error-bounded interval arithmetic and fixing incorrect results on degenerate curve-curve configurations.
- Worked directly with senior open-source maintainers through async code review cycles, refining API surface design and edge-case handling to the standards of a mature geometry library.

Inkscape

Oct 2024 – May 2025

Open Source Developer, Remote

- Independently clarified ambiguous technical requirements by reading unfamiliar code and writing minimal reproducible test cases, in a multi-hundred-thousand-line C++/GTK project with no onboarding documentation.
- Diagnosed and resolved high-impact bugs across rendering, tool interaction, and editor-state layers, shipping targeted, regression-free fixes for 3M+ active users.
- Maintained CI/CD pipeline integrity across 5+ dependent tool behaviors throughout the project's SDLC, and communicated technical decisions clearly in async review threads without direct mentorship.

Systems Limited

Jul 2023 – Sep 2023 & Jul 2024 – Sep 2024

Software Engineering Intern, Lahore, Pakistan

- Designed and deployed cloud solutions on AWS (EC2, S3, RDS, Lambda) across 3 production environments, building serverless, event-driven functions to support enterprise application workflows.
- Strengthened environment security by implementing IAM policies and bucket access controls, reducing unauthorized-access risk across all 3 environments.
- Collaborated with the Cloud App Development team, gaining hands-on training in Java and contributing to enterprise application development practices.

PROJECTS

Real-Time Healthcare AI Voice Agent | Node.js, TypeScript, Twilio, AWS Bedrock, ECS/Fargate, PostgreSQL, S3

- Shipped a production-quality, HIPAA-eligible real-time voice system integrating Twilio Media Streams with AWS Bedrock Nova Sonic, owning the architecture end-to-end through a KMS-encrypted S3 pipeline, PostgreSQL audit trails, and a full integration test suite.
- Built a config-driven, cloud-native multi-agent architecture that dynamically loads each agent's prompt, voice, locale, and call rules, supporting 20 distinct agents without code duplication.

Fieldwork | Python, Next.js, AWS (ECS Fargate, Lambda, EventBridge, SQS, DynamoDB, Cognito), AssemblyAI, Redis, Terraform

- Built an AI platform that runs adaptive voice user-interviews and synthesizes findings across respondents, with a Python orchestrator on ECS Fargate handling turn-taking and barge-in over WebSocket.
- Built an event-driven synthesis pipeline (EventBridge, SQS with DLQ, Lambda) that extracts, embeds, and synthesizes cross-interview themes, on Terraform-defined networking (custom VPC, private subnets, ALB) with CloudWatch monitoring.

SmartEngage | Python, FastAPI, LangGraph, LangChain, XGBoost, PyTorch, PostgreSQL, ChromaDB, Docker

- Built modular FastAPI services that ingest social media telemetry via async pipelines and predict content virality with a hybrid XGBoost + transformer model.
- Rebuilt the agentic RAG loop as a LangGraph StateGraph with a Postgres checkpointer over hybrid retrieval (ChromaDB dense + BM25, cross-encoder reranker), measured by a 40-question eval harness.

Vigil | Python, FastAPI, PostgreSQL, SQLAlchemy (async), Alpaca API, Pydantic, pytest/hypothesis

- Built an autonomous options-trading agent for the lablab.ai × Alpaca AI Trading Agents Hackathon, splitting responsibilities between a deterministic strategy/risk engine and an LLM portfolio manager so no model output reaches the broker unchecked.
- Designed a 12-gate deterministic risk kernel enforced as pure functions ahead of every order, with a Hypothesis property test proving the per-trade risk invariant and a 241-test suite passing under mypy --strict.

EDUCATION

Ghulam Ishaq Khan Institute

2022 – 2026

BS. Computer Science

SKILLS

Languages: C++, Python, TypeScript, JavaScript, Java, C

Frontend: React, Next.js, Tailwind CSS | **Databases:** PostgreSQL, MySQL, MongoDB, ChromaDB, DynamoDB, Redis

Backend & APIs: FastAPI, Node.js, REST API Design, Microservices, Docker, Kubernetes (familiar)

ML / AI: PyTorch, XGBoost, Scikit-learn, Transformers, LLMs, AI Agents, LangGraph, LangChain, RAG, Pandas, NumPy

Cloud & Infra: AWS (EC2, S3, RDS, Lambda, ECS/Fargate, Bedrock, SQS), Terraform | **Tools:** Git, CMake, Google Test, CI/CD, Linux